

RAVULA PREM KUMAR

Embedded R&D Engineer

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Professional Summary

Embedded R&D Engineer with 2 years and 6 months of experience in Embedded C and Linux-based systems. Skilled in C, Linux and Good knowledge log analysis, device configuration, and troubleshooting hardware-software interaction issues. Strong understanding of L2/L3 networking fundamentals and device communication protocols. Familiar with RTOS concepts, Linux environments, and structured debugging approaches.

Technical Skills

- **Programming & Automation:** C, Embedded C, WLAN Testing
- **Debugging Tools:** Wireshark, ping and iperf/iperf3, tcpdump, ping, Putty, Telnet
- **Protocols & platform:** UART, I2C, SPI, WIFI, MQTT.
- **Build & Integration:** Git, CI/CD.
- **Platforms:** Linux, Embedded System, IEEE 802.11.

Experience

RL TECH India Pvt Ltd (Client)

Embedded R&D Engineer | Mar 2025 – JAN 2026

- Worked on embedded networking device features and system-level integration.
- Performed WLAN functional validation including SSID association, authentication, DHCP IP allocation.
- Worked on L2/L3 networking protocols including VLAN, ARP, DHCP, PPPoE, TCP, and UDP, TCP/IP
- Analyzed kernel and system logs (dmesg, syslog) to identify service crashes and network instability issues.
- Worked on Wi-Fi signal strength (RSSI), channel behavior, and connectivity degradation scenarios.
- Worked on Linux-based embedded systems including user access validation, service configuration, and log analysis

Race Energy, Hyderabad

Service Engineer | Dec 2023 – Dec 2024

- Performed embedded system diagnostics and functional validation of BMS (cell voltage monitoring, temperature sensing, protection logic, fault detection).
- Analyzed CAN logs to debug subsystem communication and controller-level issues. Identified hardware interface and wiring-level faults through signal tracing and protocol-level validation.

Inno minds Software, Hyderabad

Software Engineer Trainee | Mar 2022 – Oct 2022

Project: Structural Health Monitoring (SHM) Logging & Data System

- Worked on CAN communication using Embedded C (J1939 protocol) for frame configuration and validation.
- Supported UART communication between RTLS anchor modules and CAN interfaces.
- Performed system-level testing and debugging of embedded communication modules. Validated BLE connectivity between embedded firmware and Android application

Education

- **B.Tech. Electronics & Communication Engineering (ECE)** Vaagdevi Engineering College, Warangal - 2022 61.0%
- **Intermediate** 2018 77.7%
- **SSC** 2016-85%

